

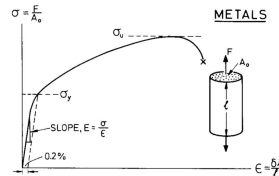
Metals and Alloys

Characterised by:

- Relatively high moduli
- Ductile, so can be formed
- Even high strength alloys, with ductility as low as 2% yields enough to ensure failure is a tough ductile type
- Partly because of ductility, prone to fatigue
- Some have poor corrosion resistance

Strengthened by:

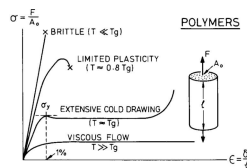
- Alloying
- Deformation
- Heat treatment



Polymers

Characteristics

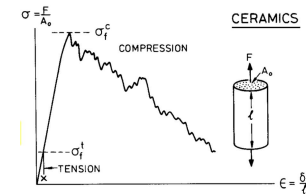
- Low moduli (~1000x less than metals)
- ~5x lighter than metals
- Some nearly as strong as metals
- Large elastic deflections
- Creep easily, even at room temp, so over time may reach permanent set if under load
- Properties depend on temperature
- NONE have useful strength above 200°C
- Easy to shape/form
- Large elastic deflections allow for snap fixtures
 - Fast & Cheap assembly
- No finishing operations needed if mould is accurate
- Corrosion resistance
- Low coefficients of friction



Ceramics & Glasses

Characteristics:

- Typically high moduli
- Brittle (No ductility)
- Strength in tension = Brittle fracture strength
- Strength in compression = Brittle crushing strength (~10x larger)
- Low tolerance for stress concentrations (holes or cracks) or high contact stresses (at clamping points)
- Scatter in strength depends upon volume of material under load
- Harder to design with than metals
- Abrasion resistant (Bearings and cutting tools)
- High temperature capability
- Corrosion resistant

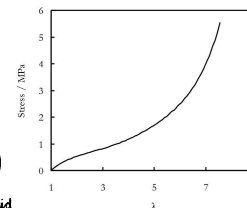


Composites

- Combination of 2 materials to exploit most attractive properties
- Typically polymer matrix (epoxy or polyester), usually reinforced by fibres such as glass, carbon fibre or Kevlar.
- Characteristics:
 - Light
 - Stiff
 - Strong
 - Tough
 - Polymer composites cannot be used above 250°C
 - Expensive
 - Difficult to form and join
 - Used if added performance justifies added cost

Elastomers

- Formed by crosslinking polymers
- Have many properties of polymers, including:
 - high extensibility
 - Low modulus (~1/100000th of metal)



4 requirements

- Temperature above glass transition temperature (T_g)
- Forces between molecules must be weak like in liquid
- Polymer must not be highly crystalline
- Polymer must be lightly crosslinked